

Answer Key — Prompt Engineering Quiz

1. B — The textual instruction, question, or example you give an AI system.
2. B — They predict the next token based on patterns learned from large text corpora.
3. A — True; tokenizers commonly split long words into sub-word tokens.
4. A — Self-attention compares and weights other tokens so representations become context-aware.
5. C — Embeddings are numeric vectors in a meaning space.
6. B — Hallucination: plausible-sounding but incorrect or fabricated information.
7. B — False; larger size helps but does not guarantee no hallucinations or errors.
8. D — All of the above (role, format, audience are useful levers).
9. A — Query/Key/Value mapping: Query = what I'm looking for; Key = what I contain; Value = what I provide.
10. B — Fine-tuning: additional training on curated examples to specialize behavior.
11. B — Emergent ability example: few-shot learning appearing in larger models.
12. A — Position embeddings encode token order since attention is permutation-invariant.
13. C — Unlimited context-window is NOT a typical capability; it's a limitation when claiming it's unlimited.
14. A — Tokenization allows handling rare or long words by splitting into subparts.
15. B — Example of effective constrained prompt is the teacher + format + length example.
16. A — Embedding metric: high-dimensional numeric vector; nearby vectors = semantic similarity.
17. A — Instruction tuning teaches the model to follow human instructions better.
18. A — Verifying facts with reliable external sources or retrieval helps reduce hallucinations.
19. A — Multi-head attention allows attending to multiple representation subspaces and capture different relationships.
20. A — Purpose, context, and constraints are the three elements good prompts typically provide.

extbfNotes for graders: Each MCQ = 1 point. For any ambiguous phrasing, use the explanation above as the authoritative key; award partial credit only if a student shows clear correct reasoning in comments.